Multi-Loss Distillation Framework

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Abstract

Although recent advances in vision models have demonstrated remarkable achievements in addressing complex visual recognition tasks, deploying these models in environments with computational constraints remains a significant challenge due to their substantial size and relatively slow inference speeds.

Adopting a similar approach to [14] for language models, a parameter-efficient alternative to conventional vision transformers has emerged, achieving near-comparable performance with significantly enhanced efficiency. Among the alternatives, TinyViT (Wu et al. [31]) stands out for its knowledge distillation technique tailored for vision transformers, which incorporates multiple loss terms across the transformer layers. Although TinyViT's performance closely matches that of its larger counterparts, a slight disparity in its empirical results persists.

This project aims to refine ViT's empirical performance by incorporating additional loss terms, as suggested by [15], to better synchronize the knowledge distillation process with the model's ability to perform visual tasks effectively. The introduction of score attributions for each visual token as an additional loss term seeks to closely align the teacher model (a larger, more complex vision transformer) with the student model.

This approach posits that optimizing the final loss of the vision model with this additional objective could bridge the performance gap between smaller models and its larger counterparts while maintaining computational efficiency. Github: https://github.com/ykamoji/multi-loss-KD

1. Introduction

While recent advancements in vision transformers have demonstrated remarkable success in image processing tasks, their deployment in computationally constrained environments, such as edge devices, poses significant challenges due to their substantial model size and slower inference speeds. To address these limitations, knowledge distillation has emerged as a promising technique to transfer capabilities from a larger model to a more compact counterpart. Originally conceptualized by [13] for neural networks, this method allows for the harnessing of a larger model's capabilities without fully utilizing its capacity during operations. Applying this technique to vision transformers, the approach involves training a smaller, less resource-intensive model to emulate the behavior of a larger one without sacrificing performance. Notable implementations of this concept include the adaptation of the vision transformer architecture into more compact forms, such as [28] and [12], which maintain competitive performance while being significantly faster and smaller. These models integrate novel distillation strategies that involve multiple loss terms across the transformer's layers to refine performance.

Although these smaller models approximate the effectiveness of their larger counterparts, a performance gap often persists. This work aims to further refine the knowledge distillation process for vision transformers by incorporating an additional loss function based on the attribution of importance across different regions of an input image. Specifically, it explores the use of this new loss function in refining the distillation process, potentially narrowing the performance gap while maintaining computational efficiency. This study builds on the ideas presented by [20], applying their concepts of attributive loss to enhance alignment between the teacher (large vision transformer) and student (compact vision transformer) models.

Our contributions are as follows:

- We adopt the GlobEnc metric to the vision domain to identify the importance of each patch existing in an image.
- We incorporate the GlobEnc metric to have a novel term for distillation loss.
- By evaluating our method on several benchmarks, we show that the proposed loss term is a reliable alternative for common distillation objectives.

1.1. Evaluations

Our evaluations includes understanding how attribution provides an effective representation, beyond what attention mechanism provides.

To ensure unbiased evaluation, we benchmark our distillation-trained models against established baseline

models pre-trained on the respective datasets. In the interest of maintaining experimental consistency and facilitating fair comparisons, we adhere to a standardized set of hyperparameters across all distillation experiments within each dataset. This methodological approach allows us to isolate the effects of our distillation technique and draw meaningful conclusions about its efficacy.

For image classification, we want to use top 1 and top 5 accuracy. For image captioning, we will use BLEU (Bilingual Evaluation Understudy) [23], ANLS (The Average Normalized Levenshtein Similarity) [2] and CAPTURE [7] metrics.

1.2. Datasets

For Image Classification, train the models on ImageNet [26]. For reduce training time, we use only 100 classes for all our experiments.

Additionally, we will further fine-tune the model for transfer learning performance evaluations using CIFAR-10, CIFAR-100 [16].

For the downstream Image Captioning task, we will use the COCO 2017 image captioning dataset [19].

2. Related Work

Currently, many SOTA models in the NLP domain use a Transformer, which has inspired many researchers in Computer Vision to bring these transformer architectures to perform a variety of vision tasks. This paper [8] introduces the concept called Vision Transformers and starting from basic classification task, segmentation [25] to multimodal application, such as MiniGPT-V2 [5].

Following the usage of transformers in vision, the researcher started exploring the method used in the NLP domain for Computer Vision. In this context, some approaches proven to work in NLP can be experimented here. One such approach is exploring Attribution-Driven Knowledge Distillation [32]. This adds to the other three approaches we discussed before. This method uses three losses to compute the final loss: a) Attribution Distillation Loss, b) KL Divergence loss and c) cross-entropy loss. Another approach [32] deals with tokens and similar paper which compress the image token using a method called Adaptive Token Sampling [9].

Moreover, there exist several approaches, including Decompx [], token-to-token interaction techniques as discussed in [10], and attention analysis methods outlined in [6], aimed at elucidating the decision-making process of a model. Some methods focus on analyzing both local and global contributions of input tokens [24]

When it comes to explainability, there are multiple level of granularity that we can look in to : 1) Local Explanation and 2) Global Explanation. Local Explanation covers the methods like Feature Attribution-Based Explanation, Attention-Based Explanation, Example-Based Explanations. The main Feature attribution-Based Explanation follows the technique called Perturbation-Based Explanation which work by perturbing input examples such as masking, altering, or/and removing input features and evaluating model output changes. The common strategy is leave-one-out that modify/removing the inputs tokens(Text) or patches (Image) at various levels that includes embedding vectors, hidden units [18], words, tokens and spans [33] to measure feature importance.

Other Attribution-Based Explanation methods includes Gradient-Based Explanation [22], Decomposition-Based Methods [3] and Surrogate Models [4]. When it comes to Attention-Based Explanation, Visualizations is most predominate method to understand the contribution of each tokens. Function-Based methods is another technique, since using only raw attention is not sufficient enough to explain model predictions completely. When comes to Image data, [9] this work introduces a parameter-free Adaptive Token Sampler (ATS) module for vision transformers, reducing computational costs by adaptively sampling significant tokens. Integrated as an additional layer along with attention layers. This seems to be useful in our work where, we not now take the attention based losses but also the ATS loss along with the attention.

Therefore, we propose to use the most recent attribution loss [21] to the self-attention layer of the DeiT-III backbone architecture [29] with a masked augmented [12] model. The intuition with the attribution loss is to give more importance to the image features among different layers in transformer that contribute the most in predicting the labels in the student model. We aim to make the student model more efficient in classifying the images. For further downstream tasks such as object segmentation, and VLMs (vision-language models) we can fine-tune this student model much faster without losing too much accuracy performance.

3. Methodology

3.1. Computing Attribution Scores

To compute attribution scores, we use the method proposed in [20] with several changes. To this end, we decompose components existing in the encoder block.

The self-attention mechanism Vaswani et al. [30] is the fundamental element of the encoder, responsible for the information mixture of a sequence of token representations $(x_1, \ldots, x_n)$. Each self-attention head computes a set of attention weights $A_h = \{\alpha_{h_{i,j}} | 1 \leq i, j \leq n\}$, where $\alpha_{h_{i,j}}$ is the raw attention weight from the ith token to the jth token in head $h \in \{1, \ldots, H\}$. Then we can write $z_i \in \mathbb{R}^d$, the output representation for the ith token of a multi-head (with

H heads), as below:

$$z_i = \text{CONCAT}(z_i^1, \ldots, z_i^H) W_O \tag{1}$$

$$z_i^h = \sum_{j=1}^{n} \alpha_{i,j}^h v^h(x_j) \tag{2}$$

where $v^h(x_j) = x_j W_v^h + b_v^h$.

Now by reformulating Equation 1, we can consider z_i as a summation over the attentions heads:

$$\begin{aligned} z_i &= \sum_{h=1}^{H} \sum_{j=1}^{n} \alpha_{i,j}^h v^h(x_j) W_O^h \\ &= \sum_{h=1}^{H} \sum_{j=1}^{n} \alpha_{i,j}^h \underbrace{v^h(x_j) W_O^h}_{f^h(x_j)} \end{aligned} \tag{3}$$

By merging W_O^h and W_v^h and discarding b_v^h, we have new z_i:

$$z_i = \sum_{j=1}^{n} \sum_{h=1}^{H} \alpha_{i,j}^h x_j W_{\text{Att}}^h \tag{4}$$

Our decomposition is expanded by incorporating layer normalization and residual connection components through the implementation of the encoder block:

$$z_i^+ = \sum_{j=1}^{n} \sum_{h=1}^{H} \alpha_{i,j}^h x_j W_{\text{Att}}^h + x_i \tag{5}$$

x_j and x_i can be written as $x_j = \sum_{k=1}^{N} x_{j \leftarrow k}$ and $x_i = \sum_{k=1}^{N} x_{i \leftarrow k}$, respectively. Now we have:

$$\begin{aligned} z_i^+ &= \sum_{h=1}^{H} \sum_{j=1}^{N} \alpha_{i,j}^h \sum_{k=1}^{N} x_{j \leftarrow k} W_{\text{Att}}^h + \sum_{k=1}^{N} x_{i \leftarrow k} \\ &= \sum_{k=1}^{N} \underbrace{\sum_{h=1}^{H} \sum_{j=1}^{N} \alpha_{i,j}^h x_{j \leftarrow k} W_{\text{Att}}^h + x_{i \leftarrow k}}_{z_{i \leftarrow k}^+} \end{aligned} \tag{6}$$

We now proceed with layer normalization:

$$\tilde{z}_i = \text{LN}(z_i^+) = \text{LN}\left(\sum_{k=1}^{N} z_{i \leftarrow k} \right) \tag{7}$$

Once more, in order to broaden the decomposition across the LN function, we utilize a method proposed by [15]. This technique involves breaking down the LN function into a summation of a new function denoted as $g(\cdot)$:

$$\text{LN}(z_i^+) = \sum_{k=1}^{N} \underbrace{g_{z_i^+}(z_{i \leftarrow k}) + \beta}_{\tilde{z}_{i \leftarrow k}} \tag{8}$$

$$g_{z_i^+}(z_{i \leftarrow k}^+) := \frac{z_{i \leftarrow k}^+ - m(z_{i \leftarrow k}^+)}{s(z_i^+)} \odot \gamma \tag{9}$$

where $m(\cdot)$ and $s(\cdot)$ correspond to the element-wise mean and standard deviation of the input vector.

By extending our decomposition, we add the layer normalization existing outside the attention block:

$$\tilde{z}_i^+ = \text{FFN}(\tilde{z}_i) + \tilde{z}_i \tag{10}$$

x_i is obtained by applying layer normalization to $\tilde{z}_i^+$:

$$\tilde{x}_i = \text{LN}(\tilde{z}_i^+) \tag{11}$$

Using the LN decomposition rule presented in Eq. 11, we separate the effects of residual and FFN output:

$$\tilde{x}_i = \sum_{k=1}^{n} \left(g_{\tilde{z}_i^+}(\text{FFN}(\tilde{z}_{i \leftarrow k})) + g_{\tilde{z}_i^+}(\tilde{z}_i^{\leftarrow k}) \right) + \beta \tag{12}$$

As a result, we obtain a representation for each token that illustrates its contribution within an encoder block:

$$\tilde{x}_{i \leftarrow k} \approx g_{\tilde{z}_i^+}(\tilde{z}_{i \leftarrow k}) = \frac{\tilde{z}_{i \leftarrow k} - m(\tilde{z}_{i \leftarrow k})}{s(\tilde{z}_i^+)} \tag{13}$$

We can now employ these representations to introduce a novel term in the loss function for knowledge distillation. We introduce A_t matrix used in 6.2 as below:

$$A_t := (||\tilde{x}_{i \leftarrow k}||) \in \mathbb{R}^{n \times n} \tag{14}$$

3.2. Knowledge Distillation

We propose to use the most recent attribution loss [21] to the self-attention layer of the DeiT-III architecture [29] and ViT models [8]. The attribution loss should give more importance to the image features among different layers in the transformer that contribute the most to predicting the labels in the student model. We aim to make the student model more efficient in classifying the images. For further downstream tasks such as image captioning, and VLMs (vision-language models), we can fine-tune this student model without losing much of the performance.

To get the best performance for our distillation, we compare the accuracy's by fine-turning the teacher model and student model separately. This is will serve as an essential reference for our experiment analysis since we expect to get better accuracy's with distillation.

$$L_{CE} = CE(\psi(Z_s), y)$$

$$L_{KL} = (1 - \lambda) CE(\psi(Z_s), y) + \lambda \tau^2 KL(\psi(\frac{Z_s}{\tau}), \psi(\frac{Z_t}{\tau}))$$

$$L_{model} = L_{CE} + L_{KL}$$

where Z_t are the logits of the teacher model, Z_s the logits of the student model, τ is the temperature for the distillation, λ is the coefficient balancing the Kullback–Leibler divergence loss (KL) and the cross-entropy (L_{CE}) on ground truth labels y, and ψ the softmax function.

Subsequently, we incorporated KL (Hard or Soft) loss alongside CE in a model supporting distillation tokens (DeiT-B/16). We then use different losses along with with CE and KL, assessing accuracy. And finally we train the model using these composite losses.

Following the [29], we use knowledge distillation from the teacher T to student S, our model, to fine-tune S. In this approach, knowledge distillation happens at both the prediction layer and intermediate layers. We incorporate the following loss function.

$$L_{\text{model}} = L_{\text{CE}} + L_{\text{KL}} + L_{\text{Attr}} + L_{\text{Attn}} + L_{\text{Hidden}}$$

We explain each term with details in what comes next.

Prediction Layer - L_{CE} and L_{KL}. Following [29], we use the cross-entropy (L_{CE}) on ground truth labels.

Intermediate Layers. We add additional loss term drawn from attribution scores:

Attribution loss:

$$L_{\text{Attr}} := \sum_{l=1}^{L} MSE(A_t^{l^T}, A_t^{l^S}) \tag{15}$$

Attention loss:

$$L_{\text{Attn}} := \sum_{l=1}^{L} MSE(A_{Attn}^{l^T}, A_{Attn}^{l^S}) \tag{16}$$

Hidden loss:

$$L_{\text{Hidden}} := \sum_{l=1}^{L} MSE(H_{Hidden}^{l^T}, H_{Hidden}^{l^S}) \tag{17}$$

To address the challenge of knowledge distillation between teacher and student models with disparate embedding and attention dimensions, we propose the incorporation of a learnable similarity loss [27] and MAD [34] (Attention Map Alignment Distillation) as an auxiliary objective. This approach aims to preserve the relational structure of the teacher's representations within the student's lower-dimensional embedding space. Thus, allowing for effective knowledge transfer through capturing the semantic structure learned by the teacher.

In addition to L_{Attr}, we also consider L_{Ats} [9] in some of our experiments:

$$L_{\text{Ats}} := \sum_{l=1}^{L} MSE(N_{ATS}^{l^T}, N_{ATS}^{l^S}) \tag{18}$$

Where

$$N_{\text{ATS}} := \left(\frac{A_{i,j} \times \|\mathbf{V}_j\|}{\sum_{k=2}^{n} A_{i,k} \times \|\mathbf{V}_k\|} \right) \in \mathbb{R}^{n \times n} \tag{19}$$

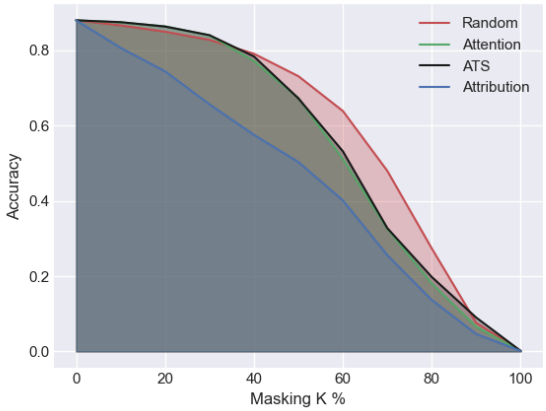


Figure 1. Area Under the Curve for Masking Accuracy

4. Experiments And Results

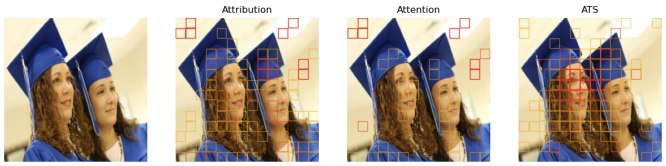
We conduct our analysis in diverse ways: 1) Evaluating score importance through visualizations, termed as faithfulness verification, 2) Examining different models on multiple datasets and exploring various loss combinations across models and datasets. When it comes to Knowledge Distillation, the most prevalent approach is use Cross Entropy and KL losses. Additionally, using the different losses mentioned in the 3.2.

4.1. Faithfulness Verification

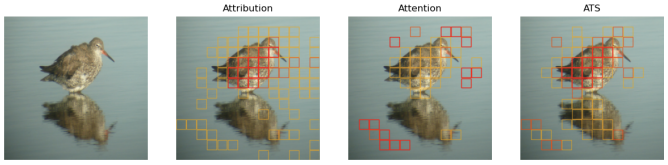
Faithfulness verification assesses whether a layer effectively captures important features of input tokens, represented by image patches. We compare three scores: Attribution, ATS, and Attention. By thresholding scores and masking important patches, we gauge the method's impact.

We randomly selected three images from a dataset (Fish, Sparrow, and Graduation) and masked them based on ATS, Attention and Attribution scores and compared them with random masking. Then, we plot the prediction accuracy concerning mask percentage and masking methods. We expect the accuracy to reduce faster with methods that represent the features well. The AUC curve is shown in 1.

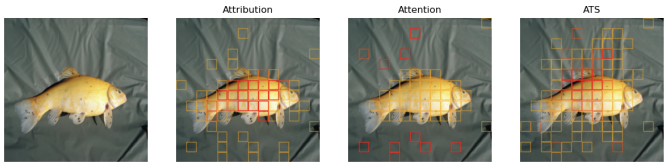
To visualize the feature scores, we use different strategies: 1) Rollout [1] that uses the representation of features by taking the dot product from every previous layer and accumulating the overall feature at the last layer of the transformer 2) Plus that uses summation of the feature across the layers 3) SkipPlus that only uses the last half layers. From



(a) Graduation



(b) Redshank



(c) Tinca tinca

Figure 2. Region scoring based on features

our experiments, we see the best representation for image classification is when plus is used. The feature importance can be seen in 2. We can observe that attribution scores perform consistently, especially in cases when even attention doesn't consider the important regions of the images as important.

Masked images are predicted by the existing model, revealing how each method prioritizes features crucial to final predictions. Our experiments involve varying levels of masking percentages (0 to 90), providing insights into the efficacy of these approaches in capturing and utilizing essential information for accurate predictions. The masking strategy can be seen in this visualization 3.

4.2. Baseline

We explore both types of distillation training of "soft" and "hard" distillation [29]. In soft distillation, the student model leverages "soft" labels from the teachers network which is the output vector of the softmax function rather than just the maximum of scores. We observe that the soft distillation improves the performance of the student model better.

We prepare a baseline performance by fine-tuning on CIFAR 10 and CIFAR 100 [16] dataset. For our experiments, we used the ViT-B/16 [8] (T) and DeiT-distilled-B/16 (S) [29] model which are pre-trained on ImageNet [26].

Our training consists of CE loss of the student outputs and ground truth labels, as well as the KL divergence soft loss between the teacher and student output logits.

The results are shown in 4. The fine-tuned ViT (T) model on CIFAR 100 has 86.15% accuracy and on the DeiT (S) model has 84.31 %. Using the more performant fine-tuned teacher model, we distill to DeiT (S), which has 85.62 % accuracy under same hyper parameters and epochs. We see an **1.31 % improvement** with distillation using CE & KL loss on the DeiT model. The training time for these two models are nearly the same.

From these experiments, we conclude that using "soft" distillation of a student model, we get better accuracy than fine-tuning the teacher model. In our further experiments, we will compare the models performance against their baseline "soft" distilled model.

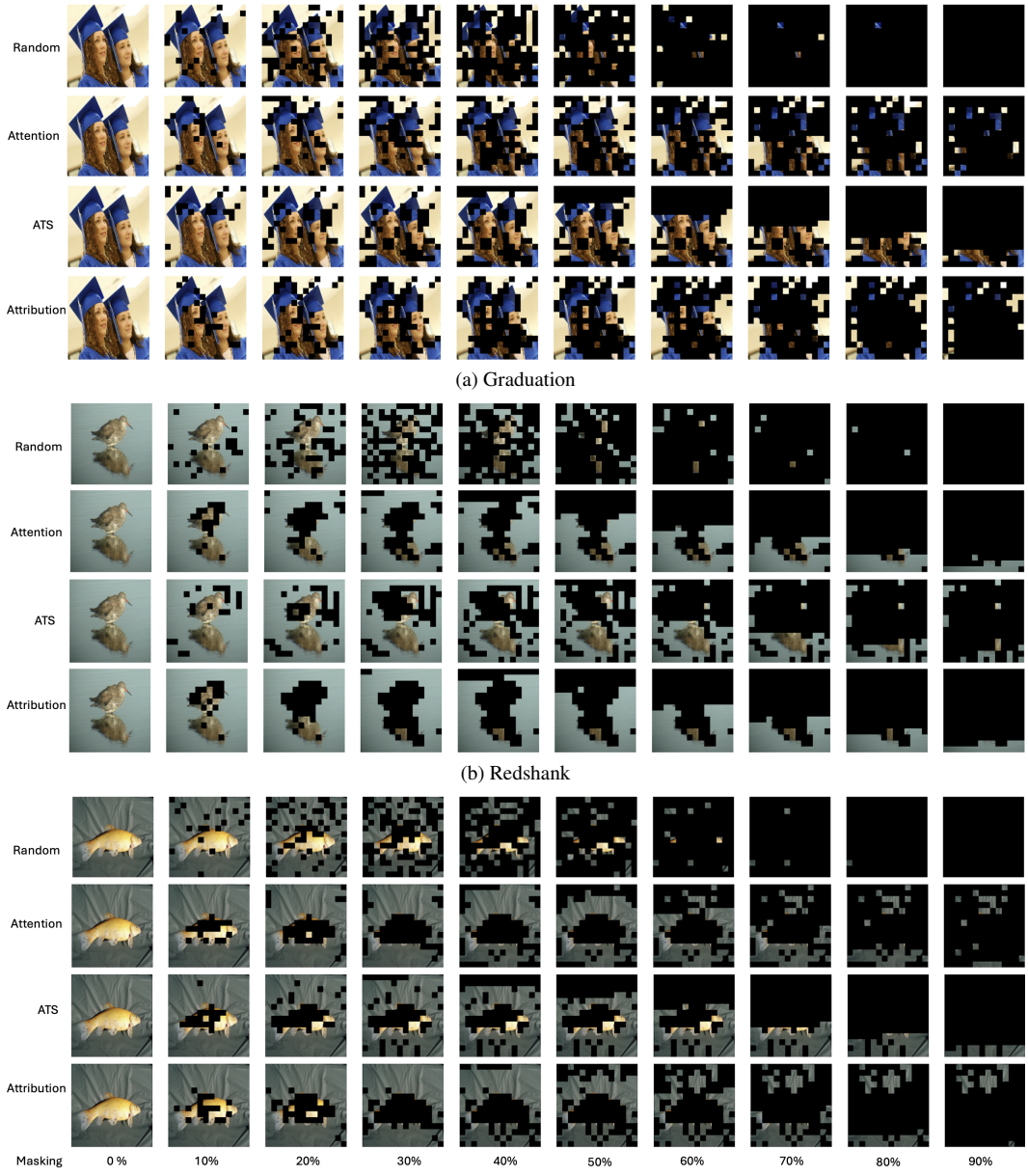
4.3. Distillation Results

4.3.1 CIFAR

Our primary aim is to optimize the training of a student model by experimenting with various loss functions and selecting the most effective combination to facilitate efficient learning from the teacher model. We sequentially introduced different losses, evaluating their impact on prediction accuracy. Initially, using a ViT-B/16 [8] model, we fine-tuned on CIFAR and assessed performance with different distillation strategies using this as the teacher model.

These experiments results are shown in 1. This shows the improvements we get on different distillation strategies. We see good accuracy improvements on both datasets. It is worth noting that the improvement for CIFAR 100 is more for CIFAR 10 even though CIFAR 100 has more labels for classification and hence a little harder.

Among the distillation strategies, adding attribution scores to the distillation loss seems to be the most effective, as it achieves better results on both CIFAR-10 and CIFAR-100. Interestingly, combining Attribution and ATS terms re-



(a) Graduation

(b) Redshank

(c) Tinca tinca

Figure 3. Patch masking based on various scores

Model	Loss Strategy	CIFAR 10	CIFAR 100	FLOPs (10^{18})	Duration (Hrs)
ViT-B/16	L_{CE}	98.21	89.34	3.871	1.542
DeiT-B/16 ⌝	L_{CE} +L_{KL}	98.43 (+0.22)	90.43 (+1.09)	3.875	1.677
DeiT-B/16 ⌝	L_{CE} + L_{KL} + L_{Ats}	98.43 (+0.22)	90.43 (+1.09)	11.649	2.211
DeiT-B/16 ⌝	L_{CE} + L_{KL} + L_{Attr}	**98.46** (**+0.25**)	**90.65** (**+1.31**)	11.652	8.836
DeiT-B/16 ⌝	L_{CE} + L_{KL} + L_{Attr} + L_{Ats}	98.29 (+0.08)	90.30 (+0.96)	11.763	7.73

Table 1. CIFAR: Fine-tuning and Distillation models of ViT & DeiT pretrained model. ⌝ donotes the DeiT model is a distilled from ViT (T).

Loss Strategy	Accuracy (top1)	Accuracy (top5)	FLOPs (10^{20})	Duration (Hrs)
L_{CE}	81.16	95.86	9.675	6.085
⌝ L_{CE} + L_{KL}	84.70 (+3.54)	97.06 (+1.2)	9.678	7.586
⌝ L_{CE} + L_{KL} +L_{Hidden^1}	84.12 (+2.96)	96.66 (+0.8)	25.892	10.29
⌝ L_{CE} + L_{KL} +L_{Hidden^2}	84.58 (+3.42)	97.24 (+1.38)	25.895	11.79
⌝ L_{CE} + L_{KL} +L_{Hidden^3}	**84.76** (+3.6)	**97.28** (+1.42)	25.903	11.51
⌝ L_{CE} + L_{KL} +L_{Attn^1}	84.42 (+3.26)	97.04 (+1.18)	22.463	7.732
⌝ L_{CE} + L_{KL} +L_{Attn^2}	84.36 (+3.2)	97.18 (+1.32)	24.625	11.67
⌝ L_{CE} + L_{KL} +L_{Attr^1}	84.53 (+3.37)	96.82 (+0.96)	67.746	55.02
⌝ L_{CE} + L_{KL} +L_{Attr^2}	84.64 (+3.48)	96.96 (+1.1)	67.774	54.25
⌝ L_{CE} + L_{KL} + L_{Hidden^3} + L_{Attr^1} + L_{Attr^2}	**84.96** (+3.8)	**97.42** (+1.56)	68.836	58.47

Table 2. ImageNet: Fine-tuning and Distillation with ViT-L/14 (T) and ViT-B/16 (S). ⌝ donotes the ViT-B/16 (S) model is a distilled from ViT-L (T).

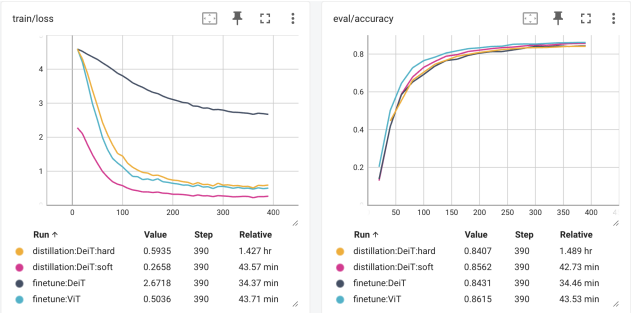


Figure 4. Baseline Models. Fine-tuning Teacher ViT and student DeiT model. Distilling from ViT (T) to DeiT (S).

sults in lower performance compared to using either of them separately. This may be because the directions of these two types of information are orthogonal to each other, leading to interference rather than synergy when combined. This observation underscores the importance of carefully selecting complementary loss components to maximize the effectiveness of distillation strategies.

4.3.2 ImageNet

For our ImageNet experiments, we used the ViT-B/16, ViT-L/16 and ViT-L/16 [29].

We performed distillation with only 100 classes of ImageNet (due to resource limitations). The results are shown in table 2 The distilled model (CE + KL) gave a higher accuracy of 84.7 % over the fine-tuned model accuracy of 81.16%.

We use various loss strategies to distill our student model.

- L_{Hidden^1}: Following the approaches in [27] and [11], we employ a similarity loss to address the differences in embedding sizes between the teacher and student models. Specifically, we compute the similarity matrices for the embeddings of both models and apply the loss directly to these matrices.
- L_{Hidden^2}: This approach follows the same logic as L_{Hidden^1}, but introduces a learnable projection layer for the student model embeddings prior to calculating the loss.
- L_{Hidden^2}: As proposed in [14], this method employs a learnable hidden classifier with an output dimension matching the teacher model's embedding dimension.
- L_{Attn^1}: This loss applies mean reduction across the attention heads to accommodate the differing number of attention heads between the student and teacher models.
- L_{Attn^2}: We integrate MAD [34], which employs a weighted attention loss that considers all attention heads.
- L_{Attr^1}: This loss is based on the direct scores outlined in 3.1.
- L_{Attr^2}: This loss follows the same logic as L_{Attr^1}, with the inclusion of a learnable projection classifier for the student model.

As shown in 2, all loss strategies incorporating additional terms to the baseline cross-entropy ($\mathcal{L}_{CE}$) and Kullback-Leibler divergence ($\mathcal{L}_{KL}$) outperform the baseline by a considerable margin in both top-1 and top-5 accuracy metrics. Among the strategies that add a single additional loss term, incorporating the hidden layer loss ($\mathcal{L}^3_{Hidden}$) achieves the highest top-1 accuracy (84.76%) and a significant gain in top-5 accuracy (+1.42%). This suggests that intermediate feature alignment, particularly deeper hidden layers, plays a crucial role in effectively mimicking the teacher model.

Interestingly, adding raw attention ($\mathcal{L}^1_{Attn}$ and $\mathcal{L}^2_{Attn}$) or attribution losses ($\mathcal{L}^1_{Attr}$ and $\mathcal{L}^2_{Attr}$) leads to a decrease in performance compared to using only $\mathcal{L}_{CE} + \mathcal{L}_{KL}$. This can be attributed to the indirect nature of the information provided by raw attention and attribution scores. These scores represent outputs that aggregate or summarize intermediate layers rather than directly aligning them. Consequently, they might introduce conflicting signals when combined with the teacher's outputs, leading to suboptimal per-

Loss Strategy	BLUE [23]	ANLS [2]	CAPTURE [7]	Duration (Hrs)
L_{CE}	0.337	0.201	0.4020	7.67
⅂ $L_{\text{CE}} + L_{\text{KL}}$	0.344 (+0.007)	0.235 (+0.034)	0.4152 (+0.0132)	7.55
⅂ $L_{\text{CE}} + L_{\text{KL}} + L_{\text{Attr}}$	0.429 (+0.092)	0.361 (+0.16)	0.4423 (+0.0403)	13.8

Table 3. COCO: Fine-tuning and Distillation of pix2struct-large (T) and pix2struct-base (S) models. ⅂ donotes the pix2struct-base (S) model is a distilled from pix2struct-large (T).

formance.

The best performance is achieved when combining all loss components ($\mathcal{L}_{CE} + \mathcal{L}_{KL} + \mathcal{L}^3_{Hidden} + \mathcal{L}^1_{Attr} + \mathcal{L}^2_{Attr}$), resulting in a top-1 accuracy of 84.96% and a top-5 accuracy of 97.42%. This indicates that while individual components may vary in effectiveness, their collective synergy provides the most comprehensive guidance for the student model during distillation. Notably, the combined approach also results in a slight increase in computational cost (68.836×10^{20} FLOPs) and training duration (58.47 hours).

These findings underscore the importance of selecting complementary loss terms for effective knowledge distillation. Specifically, leveraging direct intermediate representations ($\mathcal{L}^3_{Hidden}$) rather than indirect metrics like attention or attribution appears to be crucial for aligning the student and teacher models.

4.3.3 COCO

For this downstream task of image captioning, we use pix2struct-large (T) and pix2struct-base (S) [17] models. Due to resource constraints, we train using a smaller subset of the COCO dataset, consisting of 5,000 image-caption pairs for training and 500 images for evaluation.

The results are shown in 3. With training involving limited data and large models, the improvement in scores through distillation alone is insufficient. However, incorporating attribution loss demonstrates a more significant enhancement in the metrics.

Building on previous experiments, L_{Attr} introduce additional computations that significantly increase the training time. However, this increase can be overlooked if sufficient compute resources are available, as is typical in a multi-GPU environment.

5. Conclusion

Based on our experimental evaluation, the proposed method demonstrated its efficacy across both image classification tasks (CIFAR-10, CIFAR-100, and ImageNet) and caption generation (COCO). By incorporating score attributions as an additional loss term in the knowledge distillation process, our approach effectively enhances the alignment between the teacher and student vision transformers. This refinement led to improved empirical performance of the student models, narrowing the gap with larger counterparts without compromising computational efficiency. These findings validate the potential of our multi-loss knowledge distillation framework as a practical solution for deploying vision transformers in resource-constrained environments, paving the way for further advancements in efficient visual recognition and generation tasks.

5.1. Future works

Due to limitations on computing resources, we didn't perform experiments on the entire ImageNet and COCO datasets. However, we expect that the attribution distillation would be even more promising with more images.

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